

Session 10 Course Title: Mathematics II Course Code: MAT102/0541-102 Credits: 3	Course Teacher: Md. Yousuf Ali (YA) Lecturer, Department of GED. Email: yousufkumath@gmail.com Mobile: 01766906098
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Eigenvalues

Introduction:

- It's just a number.
- This number may be of any type: +ve, -ve, zero, integer, fraction, irrational, even imaginary, complex.
- It's associated with square matrices.
- An $n \times n$ square matrix must have n eigenvalues, neither more, nor less.
- EVs of a matrix may be (all) distinct, (all) same, (some) repeated.
- A synonym for *eigenvalue* is *characteristic value*.

An all-covering question:

$$M = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$$

Answer the following questions listed on the RHS.

1. Find the **characteristic matrix** of M
2. Find the **characteristic polynomial** of M
3. Find the **characteristic equation** of M
4. Find the **characteristic values** of M
5. Find a spectrum of M
6. Classify M
7. Find $|M|$
8. Find a spectrum of M^T
9. Find a spectrum of M^3
10. Find a spectrum of M^{-1}
11. Find a spectrum of M^{-3}
12. Find the trace of M^{-3}
13. Find $|M^{-3}|$.

This Q covers all of the information about eigenvalues.

Theory: If λ be an *eigenvalue* (or, characteristic value) of M, then

1. **Answer:** $M - \lambda I_n$ is called the characteristic matrix of M ;

Therefore Characteristic matrix of M is,

$$M - \lambda I_3 = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 4-\lambda & 6 & 6 \\ 1 & 3-\lambda & 2 \\ -1 & -4 & -3-\lambda \end{bmatrix}.$$

2. **Answer:** $|M - \lambda I_n|$ is called the characteristic polynomial of M ;

Therefore Characteristic polynomial of M is,

$$|M - \lambda I_3| = \begin{vmatrix} 4-\lambda & 6 & 6 \\ 1 & 3-\lambda & 2 \\ -1 & -4 & -3-\lambda \end{vmatrix} = -\lambda^3 + 4\lambda^2 + \lambda - 4.$$

3. **Answer:** The characteristic equation of M is obtained by **putting zero equal** to the *characteristic polynomial* ; $\therefore |M - \lambda I_n| = 0$

Therefore Characteristic equation of M is,

$$|M - \lambda I_2| = -\lambda^3 + 4\lambda^2 + \lambda - 4 = 0$$

4. **Answer:** Roots of the characteristic equation are called the eigenvalues of M ;

Therefore the eigenvalues are, (by solving the characteristic equation)

$$\therefore -\lambda^3 + 4\lambda^2 + \lambda - 4 = 0 \quad \lambda = 4, -1, 1$$

5. **Answer:** A set of eigenvalues is called a spectrum of M ;

Therefore the spectrum is,

$$\{-1, 1, 4\}$$

6. **Answer:** M is nonsingular **iff** all eigenvalues are nonzeros but M is singular **iff** any one of eigenvalues are zeroes.

Therefore

$$\lambda \neq 0 \\ \Rightarrow M \text{ is nonsingular.}$$

7. **Answer:** $|M|$ = (Product of eigenvalues of M) & $\text{tr}(M) = \Sigma(\text{eigenvalues of M})$.

Therefore

$$|M| = (-1)(1)(4) = -4$$

$$\text{tr}(M) = -1 + 1 + 4 = 4$$

8. **Answer:** M & M^T posses same spectrums.

Therefore the spectrum of M^T is,

$$\{-1, 1, 4\}$$

Theory (contd):

- 9 If M has an ev λ , then M^k has an ev λ^k , when $k = 2, 3, \dots, r$;
- 10 If M has ev λ & M is nonsingular, then $1/\lambda$ is an ev of M^{-1} .
- 11 The result follows when 9 & 10 are combined.

Solution (in progress):

- 9 Here $k = 3$ & $\lambda = 4, -1, 1$
 $\Rightarrow M^3$ has evs $\lambda^3 = 4^3, (-1)^3, 1^3$
 $\Rightarrow M^3$ has a spectrum $\{64, -1, 1\}$.
- 10 M is nonsingular [see 6 at sld-7].
 $\therefore M$ has ev : $4, -1, 1$
 $\therefore$ spectrum of M^{-1} is $\{1/4, -1, 1\}$.
- 11 A spectrum of M^{-3} is $\{1/64, -1, 1\}$.

12 $\text{tr}(M^{-3}) = (1/64) + 1 + (-1) = 1/64 .$

13 $|M^{-3}| = (1/64) (1) (-1) = -1/64 .$

SQ

- How many evs may a 13×13 matrix have ?
- How many evs do belong to a 17×18 matrix ?
- A singular matrix as a spectrum $\{-4, 1, 7\}$: **True or False ?**
- A nonsingular matrix must have only nonzero evs : **True or False ?**
- Since $\text{tr}(M) = \Sigma(\text{evs of } M)$, then evs of M are diagonal elements of M : **True or False ?**

BQ

- Answer all questions mentioned in the sld-3, with respect to the following matrices:

- $\begin{pmatrix} 4 & 3 \\ 2 & 5 \end{pmatrix}, \begin{pmatrix} 1 & 4 \\ 9 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix}$
- $\begin{pmatrix} 5 & 7 & 1 \\ 0 & -2 & 6 \\ 0 & 0 & 3 \end{pmatrix}, \begin{pmatrix} 6 & 3 & -3 \\ -2 & -1 & 2 \\ 16 & 8 & -7 \end{pmatrix}, \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 1 \\ 2 & 1 & 0 \end{pmatrix}$
- $\begin{pmatrix} 5 & 0 & 4 \\ 0 & 2 & 1 \\ 4 & 1 & 3 \end{pmatrix}, \begin{pmatrix} 3 & 4 & 0 \\ -4 & 5 & 9 \\ 0 & 8 & 6 \end{pmatrix}$

- In some cases, your answer may be undefined.
- In some cases c-numbers may occur as answers.